

Multimedia Application

By

Minhaz Uddin Ahmed, PhD

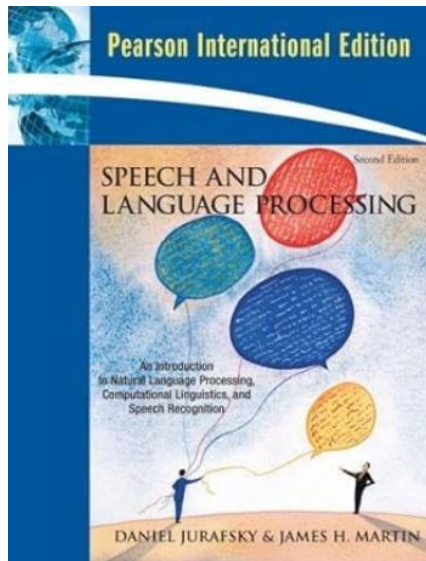
Department of Computer Engineering

Inha University in Tashkent.

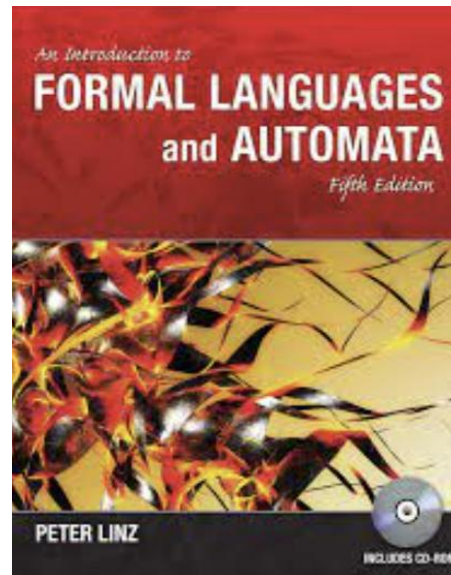
Email: minhaz.ahmed@gmail.com

Course Overview

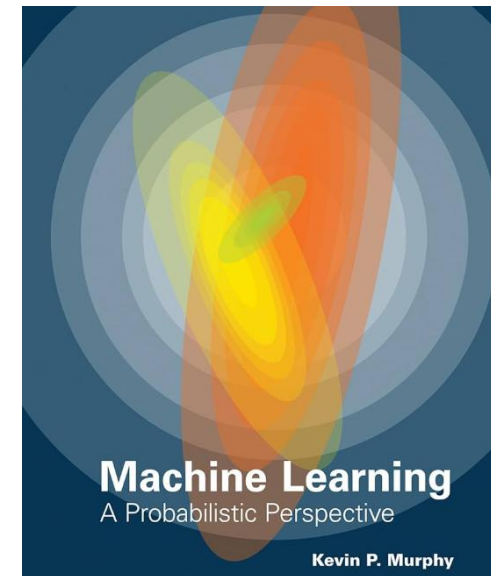
► Textbook



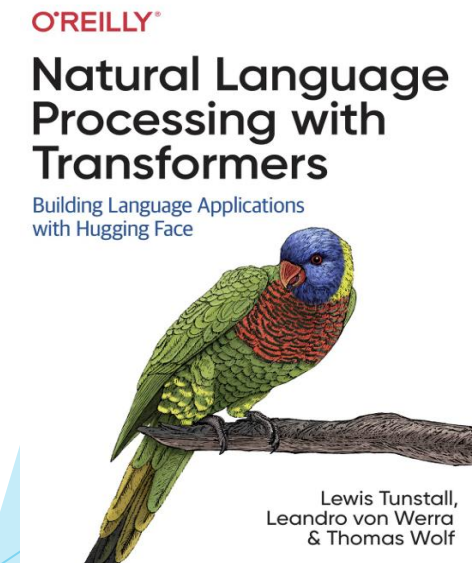
Speech and Language Processing



Formal languages and automata



Advances in ML



Transformers

Course overview

▶ Course Evaluation:

- ▶ Mid term Exam 30%
- ▶ Final Exam 30%
- ▶ Assignment 15%
- ▶ Quiz 20%
- ▶ Attendance & active participation 5%

Assignments are based on python programming

Course Syllabus

Week	Content
Week One	Introduction to Multimedia Application
Week Two	Basic Text Processing ○ Text and speech basic
Week Three	Basic Text Processing ○ Text Normalization ○ Understanding how words and sentences are structured
Week Four	Fundamental algorithms for NLP ○ Regular Expressions, ○ Neural Language Models
Week Five	Fundamental algorithms for NLP ○ Naive Bayes, ○ Text Classification, and Sentiment

Course Syllabus

Week	Content
Week six	Part-of-Speech Tagging ○ Identifying the grammatical role of each word (e.g., noun, verb, adjective).
Week Seven	Parsing ○ constituency parsing, ○ dependency parsing.
Week Eight	Mid-Term Exam
Week Nine	NLP application Tools ○ NLTK, spaCy ○ Transformers
Week Ten	NLP application ○ Machine Translation ○ Question Answering and Information Retrieval

Course Syllabus

Week	Content
Week Eleven	NLP dataset ○ Large collections of text data used for training NLP models.
Week Twelve	Deep Learning for NLP ○ Using neural networks for various NLP tasks. ○ Word embeddings ○ representing words as vectors
Week Thirteen	Deep Learning for NLP ○ Recurrent Neural Networks (RNNs) ○ Transformers ○ Attention-based models
Week Fourteen	Recent Advances in NLP Research ○ Conversational AI ○ Multimodal AI (Visual and Textual Understanding) ○ Text and Speech Integration ○ Ethical Considerations and Bias Reduction
Week Fifteen	Final Exam
Week Sixteen	Review Class

Introduction to multimedia

- ▶ Multimedia is media and contents that uses a combination of different content forms.
- ▶ Multimedia includes a combination of
 - ▶ Text
 - ▶ Audio
 - ▶ Still Images
 - ▶ Animation Video
 - ▶ Interactivity content forms



Introduction to multimedia

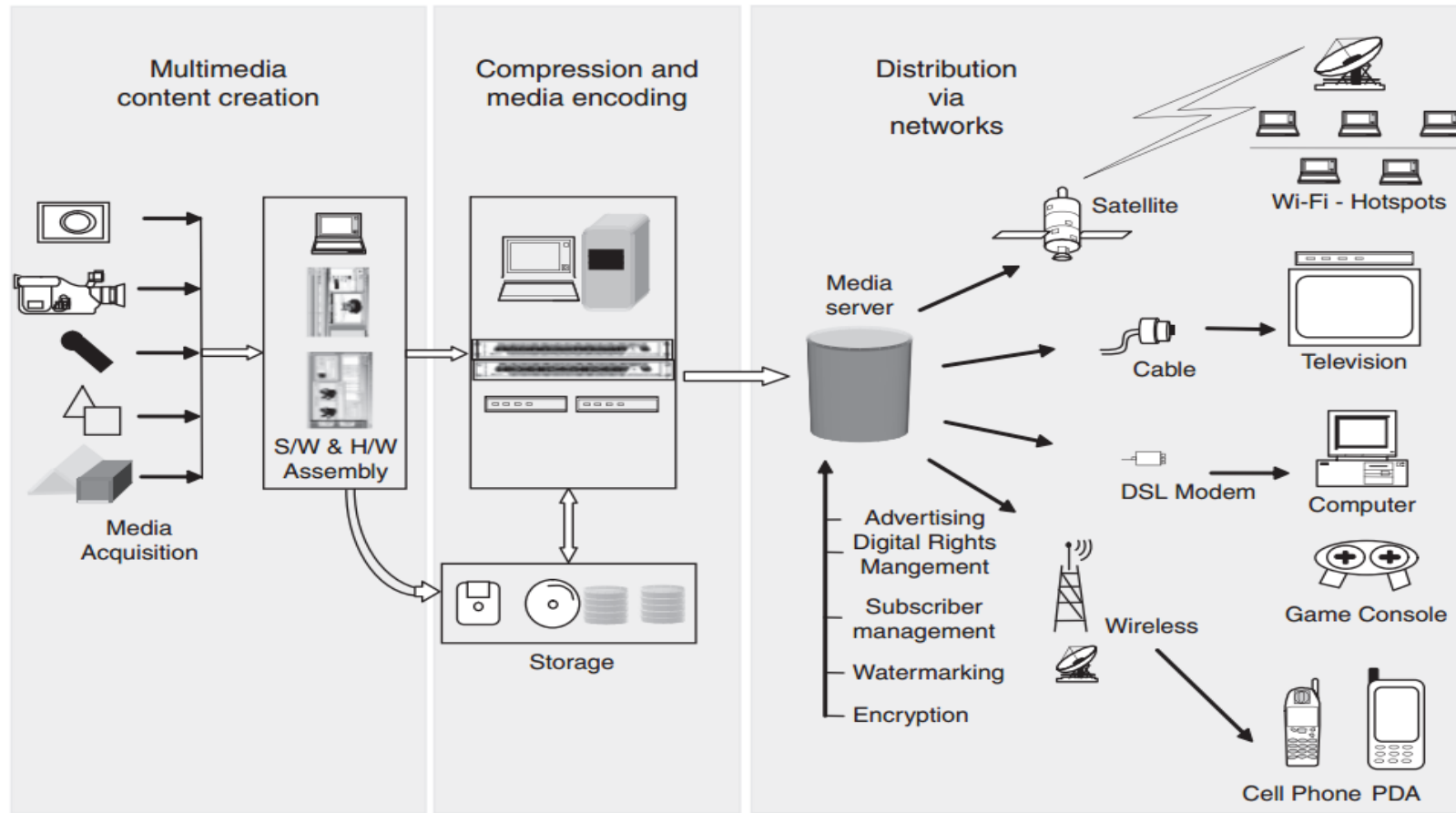


Figure 1-2 Components of a multimedia system today

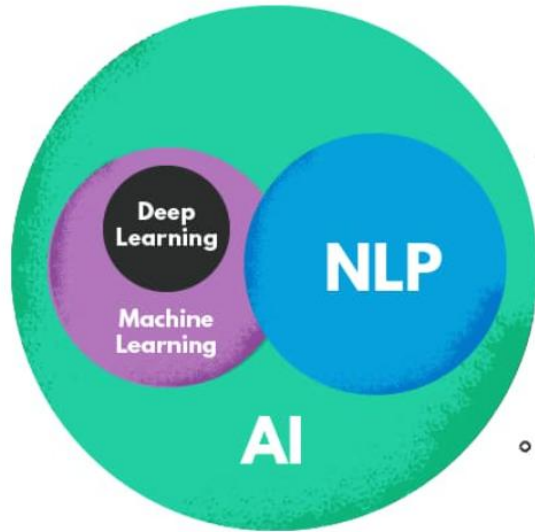
Introduction to multimedia

1. A PowerPoint presentation. Yes—95%. PowerPoint presentations involve all types of media and have tools to make it interactive.
2. Playing a video game. Yes—100%. Video games are inherently interactive.
3. Describing a picture to your friend. Yes—10%.
4. Reading the newspaper in the morning. Yes—20%.
5. Videoconferencing. Yes—100%.
6. Watching television or listening to radio. Yes—80%. Most said yes because TV comprises audio and video, with channel surfing as interactivity.

Introduction to multimedia computing

- ▶ TEXT
- ▶ Text is the most widely used and flexible means of presenting information on screen and conveying ideas.
- ▶ Text is an essential aspect of presenting the information.
- ▶ Like each element of the multimedia design, effective use of text can either direct users/readers attention or divert it.

Natural Language Processing



- ▶ NLP is a subfield of (AI) that focuses on the interaction between computers and human language. It involves the development of algorithms and models to enable computers to understand, interpret, and generate human language in a meaningful way. NLP plays a crucial role in various applications, including machine translation, sentiment analysis, chatbots, information extraction, and text summarization.

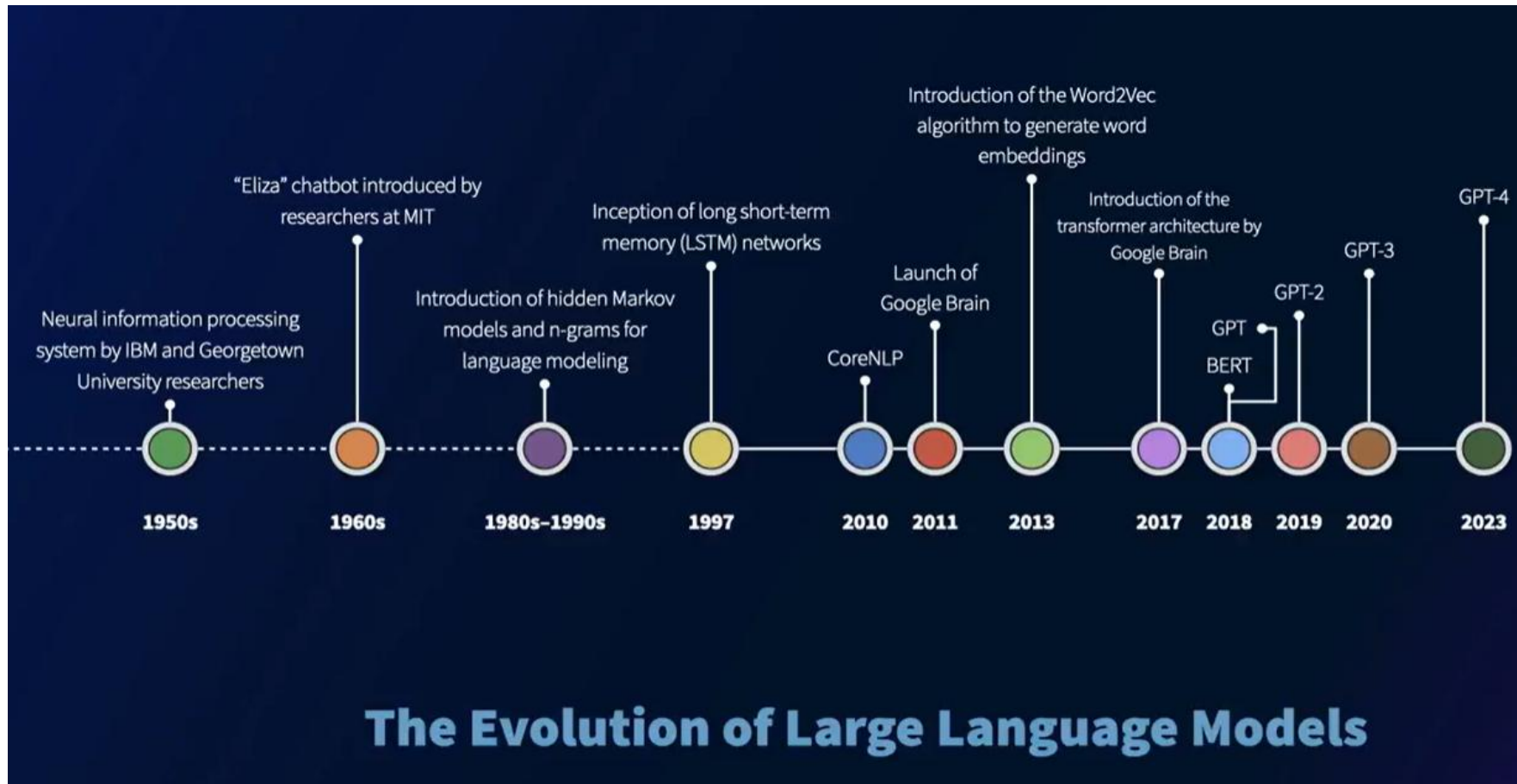
Natural Language Processing

- ▶ **Tokenization:** Tokenization is the process of breaking text into smaller units, such as words or subwords, known as tokens.
- ▶ **Text Preprocessing:** Text preprocessing involves cleaning and standardizing text data to improve the quality of subsequent NLP tasks
- ▶ **Word Embeddings:** Word embeddings are dense vector representations of words in a high-dimensional space, where words with similar meanings are closer together. Popular word embedding techniques include Word2Vec, GloVe, and fastText etc.

Natural Language Processing

- ▶ **Sentiment Analysis:** Sentiment analysis, also known as opinion mining, is the task of determining the sentiment or opinion expressed in a piece of text, such as positive, negative, or neutral. Sentiment analysis models are used in social media monitoring, customer feedback analysis, and brand reputation management.
- ▶ **Machine Translation:** Machine translation involves the automatic translation of text from one language to another. Statistical machine translation (SMT) and neural machine translation (NMT) are two common approaches used in machine translation systems.
- ▶ **Text Generation:** Text generation is the task of generating human-like text based on a given prompt or context.

Natural Language Processing



Images

- ▶ Pictures(images) is a two-dimensional screen display, and as well as a three-dimensional, such as a statue or hologram.
- ▶ Graphs, pie-charts, painting etc. all come under images.
- ▶ Images are a very useful feature of multimedia. Multimedia presentation uses pictures or clip-art to make people understand.
- ▶ Various file formats of images are .jpg, .png, .gif etc



AUDIO

- ▶ Audio refers to sound. Multimedia can include files which contain sounds
- ▶ Audio songs also come under the heading multimedia
- ▶ Multimedia presentations often have some audio tracks which makes it easier for people to understand.
- ▶ Multimedia phones have music players to run audio music
- ▶ Various audio software include VLC media player, real player, etc

AUDIO

- ▶ Audio Compression: Audio compression algorithms are used to reduce the file size of audio recordings while maintaining acceptable sound quality. Lossy compression formats such as MP3 and AAC discard some audio data to achieve higher compression ratios, while lossless compression formats such as FLAC preserve all audio data without loss of quality.



Audio Synthesis



- ▶ **Audio Synthesis:** Audio synthesis involves generating sound electronically using oscillators, filters, envelopes, and modulation techniques. Synthesizers, samplers, and electronic music production software enable musicians and sound designers to create a wide range of sounds and musical textures.

Voice Recognition and Synthesis

- ▶ Voice recognition technology enables computers to interpret and respond to spoken commands, while text-to-speech (TTS) synthesis technology converts written text into spoken words. These technologies are used in virtual assistants, automated phone systems, navigation systems, and accessibility tools.



VIDEO

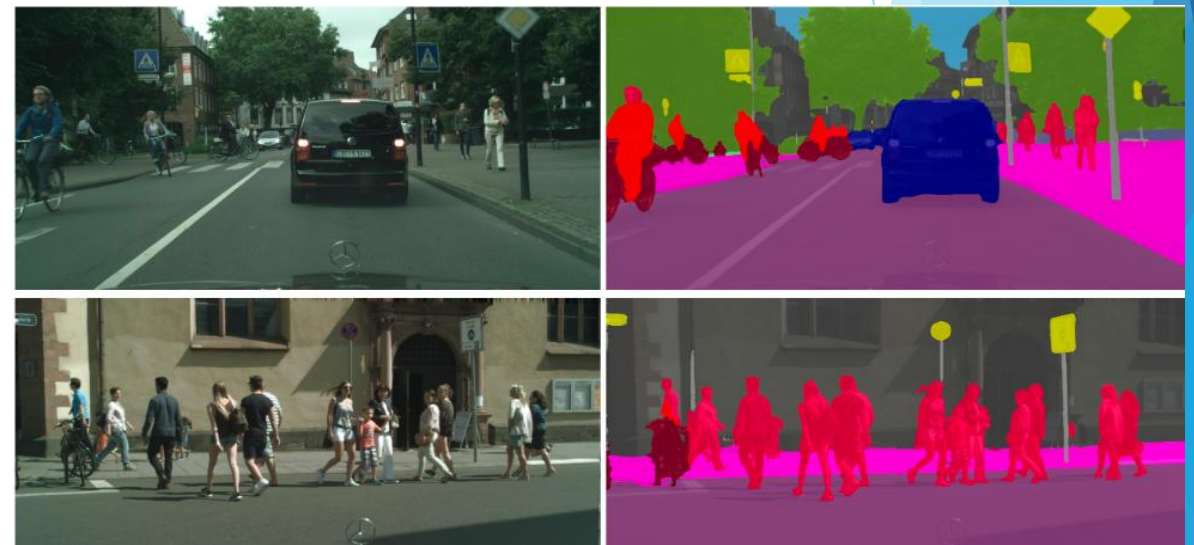
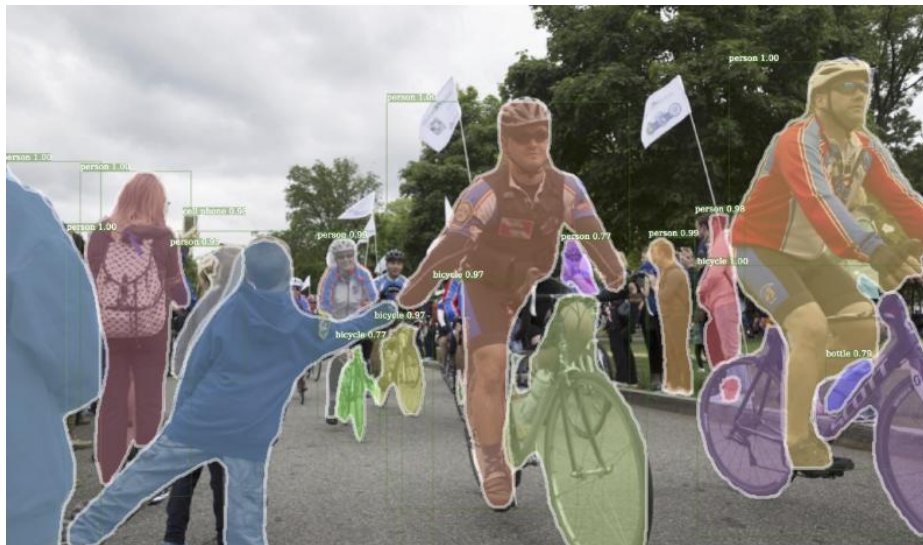
- ▶ A video is unedited material as it had been originally filmed by movie camera or recorded by a video camera.
- ▶ The embedding of video in multimedia applications is a powerful way to convey information which can incorporate a personal element which other media lack.
- ▶ Video enhances, dramatizes, and gives impact to your multimedia application.
- ▶ The advantage of integrating video into a multimedia presentation is the capacity to effectively convey a great deal of information in the least amount of time

Applications of digital image processing

- ▶ Necessity of digital image processing
 - improvement of pictorial information for human perception
 - image processing for autonomous machine application
 - Efficient storage and transmission

Image processing applied to computer vision

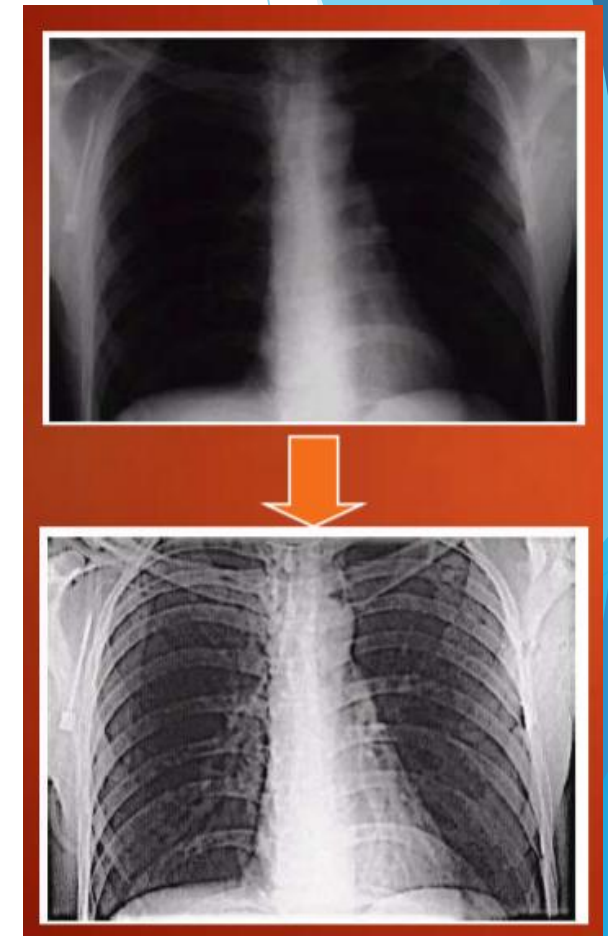
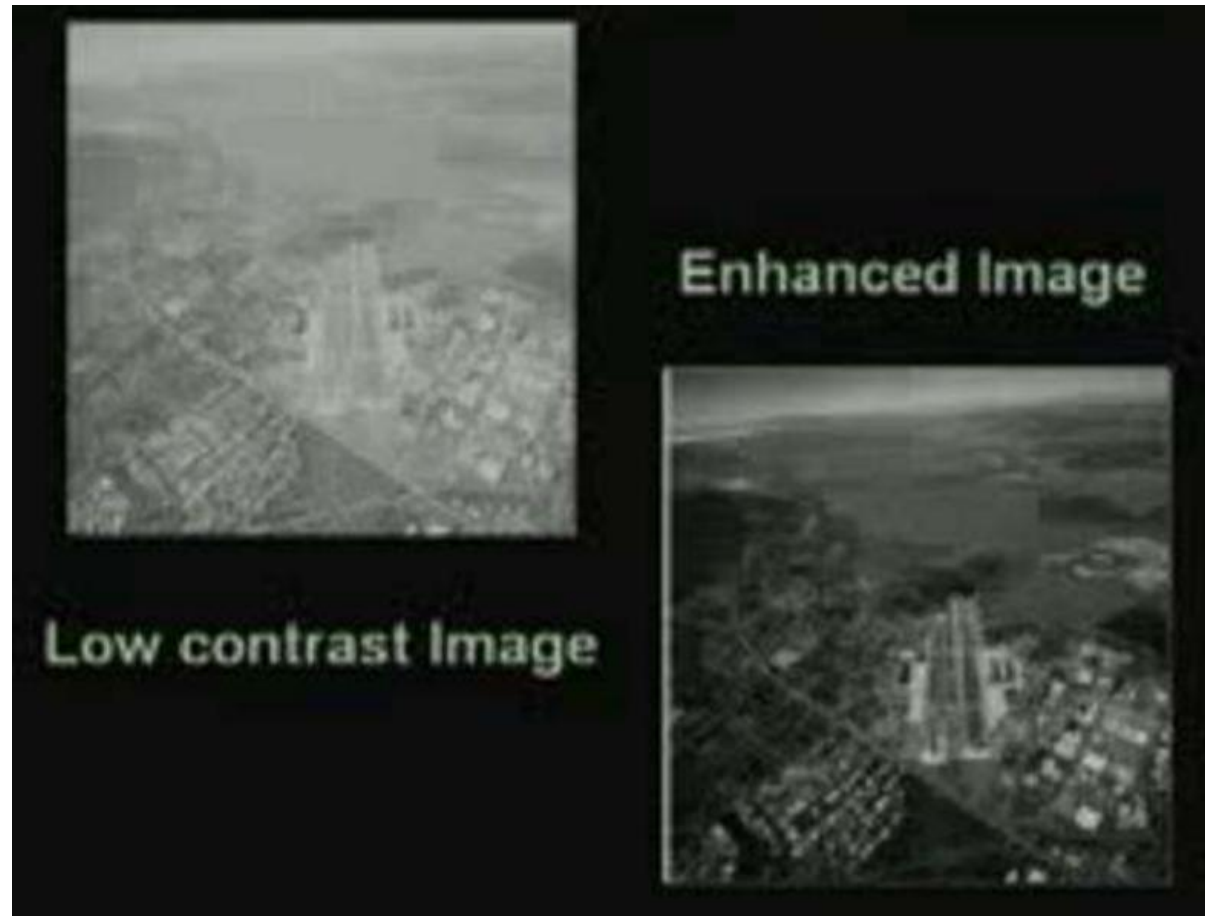
- ▶ Low level (Noise removal, image sharpening)
- ▶ Mid level (object segmentation)
- ▶ High level (Scene understanding)



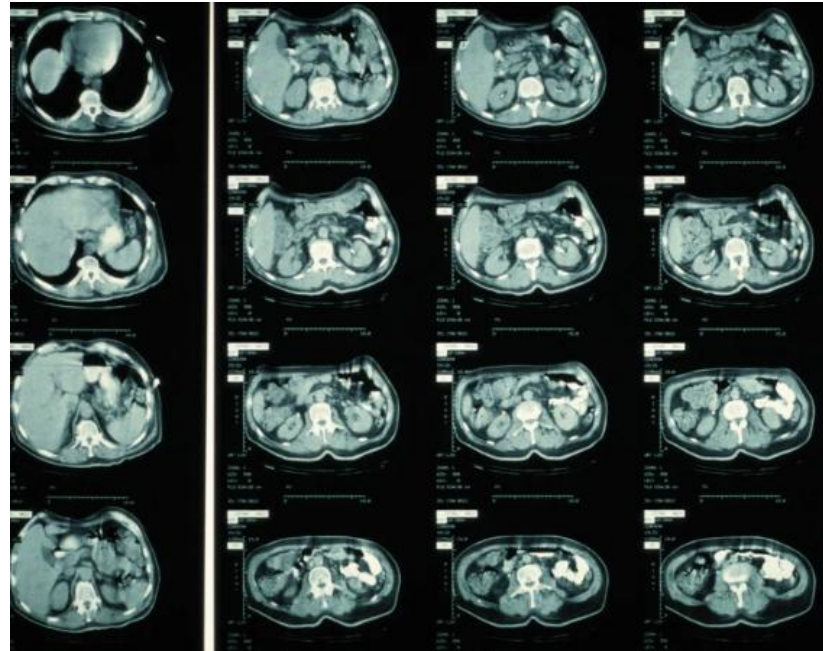
Improvement of pictorial information for human perception

- ▶ Noise filtering
- ▶ Content enhancement
 - ▶ Contrast enhancement
 - ▶ Deblurring
- ▶ Remote sensing

Image enhancement



Computed Tomography (CT)



- ▶ Computed Tomography (CT): CT imaging, also referred to as CAT scan, utilizes X-rays and computer processing to generate cross-sectional images (slices) of the body. CT scans provide detailed images of bones, soft tissues, and organs, making them valuable for diagnosing conditions such as tumors, fractures, and internal injuries.

Magnetic Resonance Imaging (MRI)



Magnetic Resonance Imaging (MRI): MRI uses strong magnetic fields and radio waves to produce detailed images of the body's internal structures. MRI is particularly useful for imaging soft tissues, such as the brain, spinal cord, muscles, and joints. It is commonly used in neuroimaging, musculoskeletal imaging, and oncology.

Ultrasound imaging

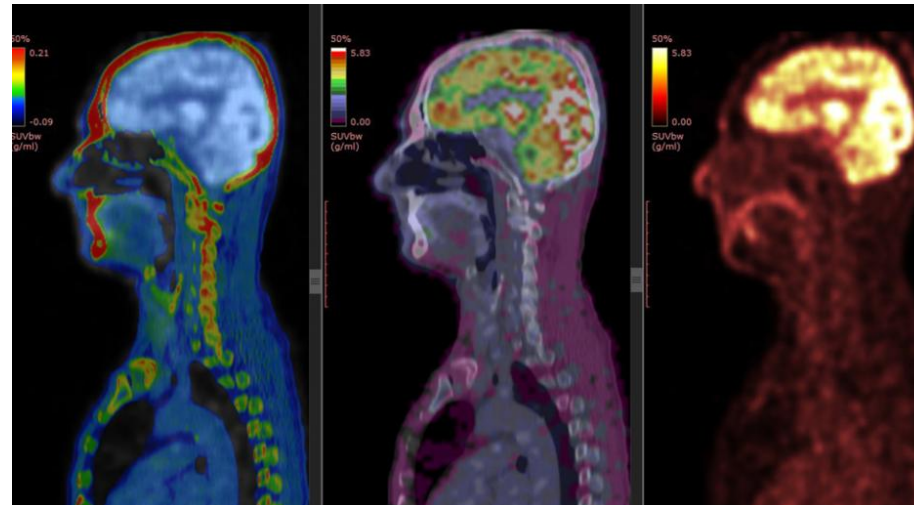


Ultrasound imaging (sonography) is a diagnostic medical procedure that uses high-frequency sound waves to produce dynamic visual images of organs, tissues or blood flow inside the body. Ultrasound is based on sonar, and uses a machine with a computer processor to create Ultrasound images.

Nuclear Medicine Imaging

- ▶ Nuclear Medicine Imaging: Nuclear medicine imaging techniques, such as positron emission tomography (PET) and single-photon emission computed tomography (SPECT), involve the use of radioactive tracers (radiopharmaceuticals) to visualize and assess the function of organs and tissues within the body. PET scans are used for cancer diagnosis, staging, and treatment planning, while SPECT scans are used for imaging the brain, heart, and bones.

- Bone Scan
- Cardiac Stress Tests
- GI Bleed Scan
- Hepatobiliary Scan
- Lung Ventilation and Perfusion Scan



Industrial inspection

- Human operators are expensive, slow and unreliable
- Make machines do the job instead

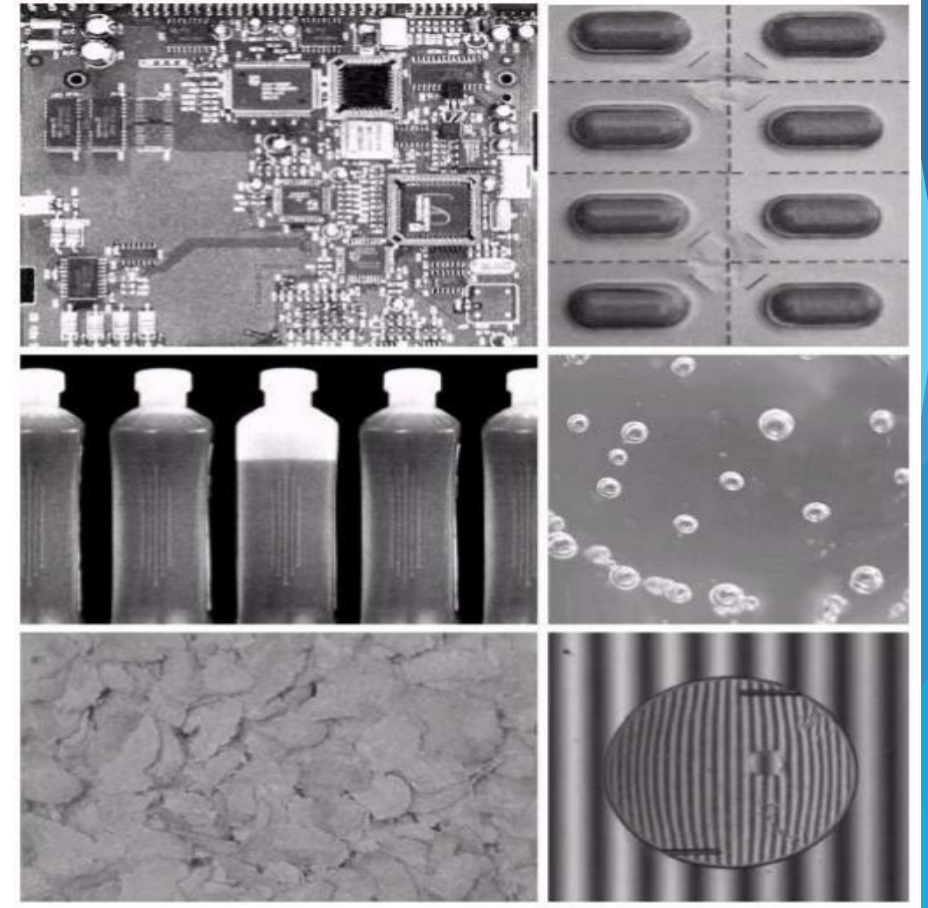


Image processing used by Law Enforcement



Car number plate



Finger print

Special effects

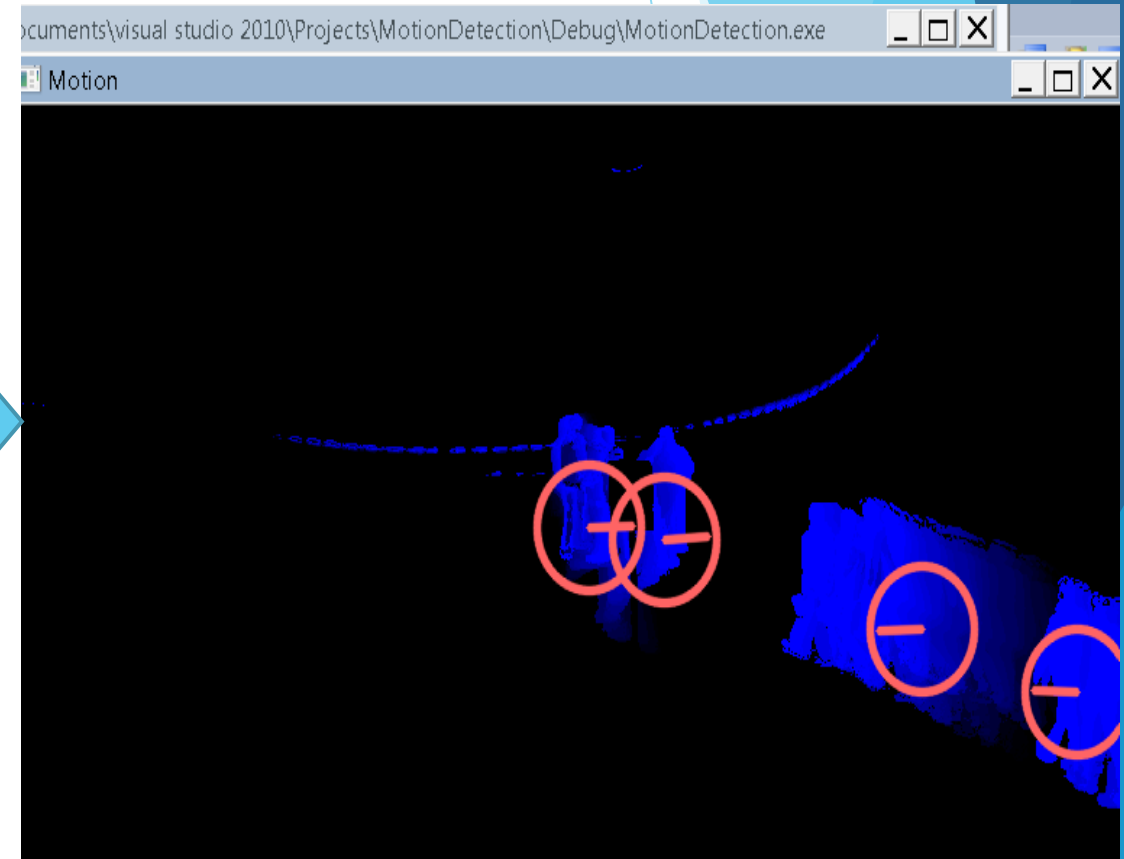
- ▶ Artistic effects are used to make images more visually appealing, to add special effects and to make composite images



Motion detection

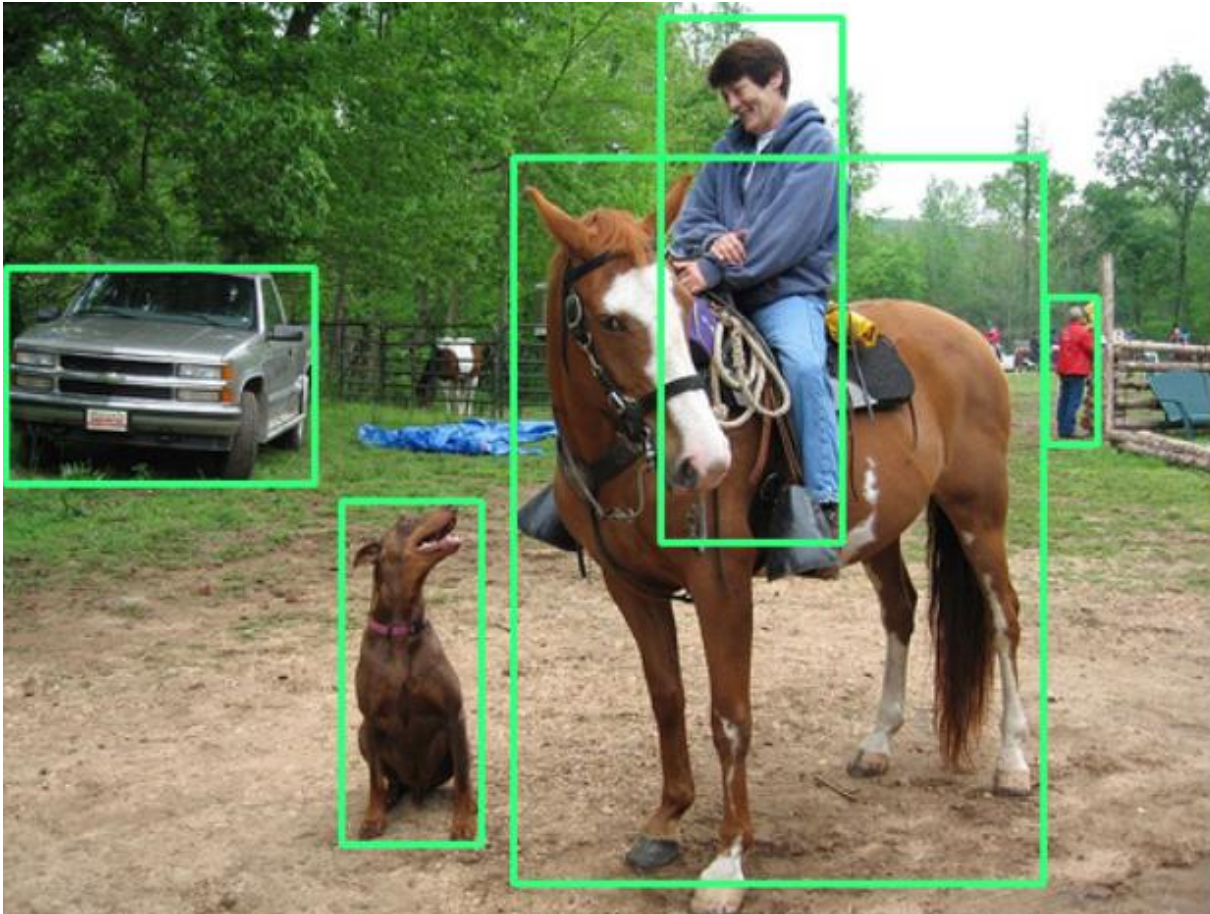


Input video



Detected motion

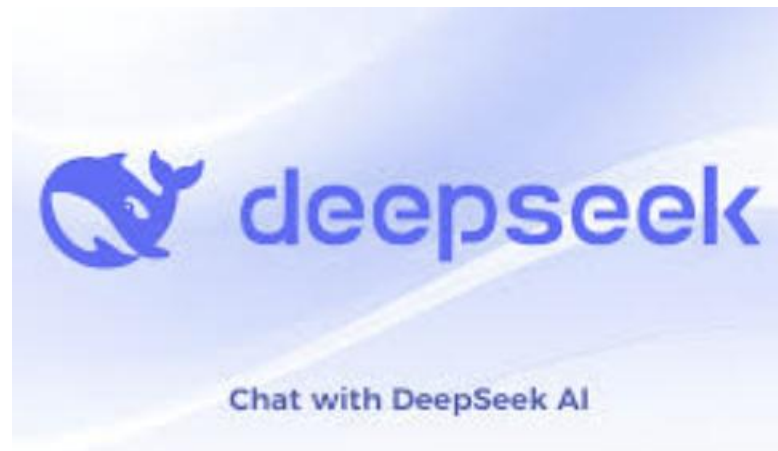
Object detection



Computer Vision is a field of computer science that works on **enabling computers** to see, identify and process images in the same way that human vision does and then provide appropriate output.

NLP Application area

- ▶ Chatbots and Virtual Assistants



NLP Application area

- ▶ Personal Assistants (Alexa, Siri)



Siri - Apple



NLP Application area

- ▶ Language Translation



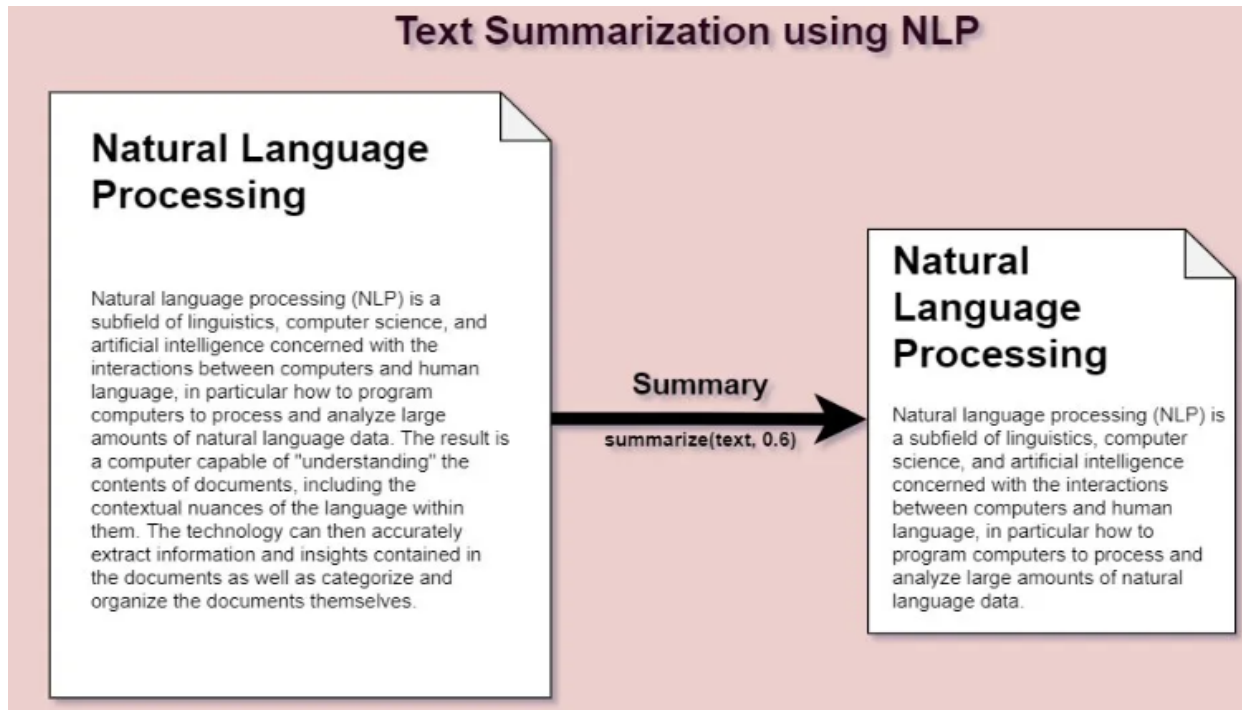
NLP Application area

- ▶ Sentiment Analysis (brand monitoring)



NLP Application area

► Text Summarization



NLP Application area

- ▶ Information Extraction
- ▶ Knowledge management



NLP Application area

- ▶ Content generation
- ▶ Content Moderation



NLP Application area

- ▶ **Personalized Search:** Search engines use NLP to personalize search results based on user preferences and past behavior.
- ▶ **Medical Records:** NLP can analyze medical records to extract information, identify patterns, and improve patient care.
- ▶ **Drug Discovery:** NLP is used to analyze scientific literature and identify potential drug candidates



Reference

- ▶ Digital image processing - Rafael C gonzalez
- ▶ Speech and Language Processing: International Edition -Daniel Jurafsky, James H. Martin

Question



Thank you

